Diagnosing Candidate-Set and Reranking Failures in Scientific Claim Retrieval: A Controlled SciFact Case Study

Saqlain Ahmed
Independent Researcher
saqlainjuna@gmail.com

August 17, 2026

Abstract

Retrieval pipelines are usually evaluated with aggregate ranking metrics, which show how often a pipeline succeeds but not why it fails. A pipeline can fail in two ways: the first stage never retrieves the relevant document, or it does but the reranker does not rank it highly. We study this distinction on SciFact by labelling every failed query as one of these two cases. We evaluate three retrievers — BM25, a BGE bi-encoder, and their reciprocal rank fusion — each followed by the same MS MARCO cross-encoder; all models are frozen. Our results show that fusion gives the best candidate recall, but final top-10 success increases only from 84.6% to 87.7%. The main change is in the type of failure: the fraction of failures caused by the first stage drops from 76% to 25%, while reranking failures become more common. We also find that the cross-encoder does not significantly improve ranking quality on the held-out split. A logistic model trained on features from the retrieval output separates failing queries better than the raw reranker score, but as pre-specified is significantly worse calibrated, so it is more useful for abstention and routing than for reporting confidence; a post-hoc ablation attributes that miscalibration to the pre-specified class weighting, which cost probability quality without buying discrimination. Overall, better first-stage recall did not lead to a comparable improvement in final top-10 success, and a larger share of the remaining failures occurred at the reranking stage. Our findings apply to SciFact and the models we evaluate.

Keywords: information retrieval; failure analysis; scientific claim verification; cross-encoder reranking; reciprocal rank fusion; query performance prediction; selective prediction; probability calibration; SciFact.

1 Introduction

A two-stage retrieval pipeline can fail in two structurally different ways. Either the first stage never surfaces the relevant document, in which case the second-stage reranker is irrelevant to the outcome, or the first stage surfaces it and the reranker fails to place it near the top. The two situations call for opposite remedies: more candidate recall in the first case, a better reranker in the second. A single Recall@10 or nDCG@10 figure does not distinguish them, so two systems with the same headline number can sit one improvement away from very different ceilings.

We study that distinction on SciFact [23], a scientific claim-verification dataset whose retrieval task is to find the abstract containing the annotated evidence for a claim. Three questions organize the work:

1. How do lexical, dense and hybrid first-stage retrieval compare on candidate recall under an otherwise identical pipeline?

2. When a pipeline fails, is the failure a candidate-set failure or a reranking failure, and does that composition change as the first stage improves?
3. Can a lightweight model, using only signals available at inference time, predict which queries will fail better than the reranker's own top score?

The study is narrow by design. No model is fine-tuned, no additional dataset is introduced, and we do not try to build a stronger retriever. The contribution is a decomposition, a protocol that keeps its parts separable, and a full accounting of the results, two of which came out contrary to our working hypothesis.

Contributions.

1. An exhaustive decomposition of multi-stage retrieval failures into *candidate-set failures* and *reranking failures*, via a five-way transition taxonomy that partitions the query set exactly (Section 3.4).
2. A controlled comparison of BM25, a frozen dense bi-encoder and their reciprocal rank fusion under a shared candidate depth and a shared frozen reranker (Section 4.1).
3. Evidence that, on SciFact, improving candidate recall shifts the composition of the remaining failures toward the reranking stage without an equally large improvement in final success (Section 4.2).
4. An evaluation of a frozen general-domain MS MARCO cross-encoder on scientific claims reporting only what the data supports: reranking does not reliably improve final ranking quality here, and a stronger calibration-set claim of degradation did not replicate held out (Sections 4.3, 4.5).
5. A lightweight inference-time failure-risk model that improves discrimination over the raw reranker score while, as pre-specified, producing significantly worse probability calibration — two properties usually reported as one — together with a post-hoc ablation attributing that calibration loss to the pre-specified class weighting rather than to the feature set (Section 4.4).
6. A leakage-conscious process in which the SciFact test split was opened exactly once, after models, features, thresholds and the primary-pipeline rule were fixed (Sections 3.7, 9).

Terminology. We use *failure-risk model* (equivalently *risk model* or *lightweight confidence model*) for the logistic model of Section 3.5, and reserve *calibration* for probability calibration in the Brier-score and reliability-diagram sense. The released artifacts name this model `common-feature calibrator`; that name is retained in the repository and in Figure 3 so that every number in this paper remains traceable to its source file.

2 Related Work

SciFact, scientific document retrieval and zero-shot evaluation. SciFact frames scientific claim verification as retrieving abstracts that contain evidence for or against a claim, together with annotated rationales [23]. Two properties make it a useful setting here: relevance is defined by *annotated* evidence rather than by topicality, which sharpens what a retrieval failure means, and an abstract that never enters the candidate set cannot be recovered downstream, which is the structural asymmetry the decomposition exploits. Scientific retrieval has motivated domain-specific encoders such as SPECTER [7]; we use general-domain models throughout and measure domain transfer instead of engineering around it. SciFact is accessed through BEIR, a heterogeneous benchmark for zero-shot retrieval evaluation [22], whose central finding is that no single paradigm dominates every collection: reranking pipelines transfer well on average while BM25 remains competitive on individual datasets. Sections 4.3 and 4.5 are a dataset-specific instance of that variance, not a contradiction of it.

Sparse, dense and hybrid retrieval. BM25 remains the standard term-matching baseline [20]; we use the `bm25s` implementation [14]. Dense bi-encoders embed queries and documents independently and match in vector space [13]; the encoder used here is trained with the C-Pack recipe [24]. The two families fail differently, one on lexical mismatch and the other on embedding drift, so combining them is common practice. Reciprocal Rank Fusion (RRF) does so using rank positions instead of raw scores, which spares us having to reconcile the incompatible scales of BM25 and cosine similarity [8]. We use it unmodified and untuned, as a fixed comparator.

Cross-encoder reranking and domain transfer. Cross-encoders score a query and a document jointly and are the standard second stage [17], with sequence-to-sequence variants following the same design [18]. Such models are typically trained on MS MARCO [3], a general-domain web collection. Section 4.3 measures how one such reranker transfers to scientific claims. Because the reranker is frozen, the study observes the interaction between candidate recall and reranking instead of training a new end-to-end system.

Query performance prediction, selective prediction and calibration. Query performance prediction (QPP) estimates how well a system has done on a query from the retrieval output alone, without relevance judgments; the pre- and post-retrieval families are surveyed by Carmel and Yom-Tov [5], with clarity-based [9] and score-distribution-based [21] predictors as canonical instances. Recent work extends QPP to conversational search [15] and documents its limits: Chifu et al. [6] report that predictor accuracy varies strongly with collection and ranker and that QPP-driven selective query processing yields only marginal gains. Our features are in the post-retrieval score-distribution tradition, but the target differs: we predict the binary event that the final top ten contains annotated evidence, which is the quantity an abstention policy consumes, and not a graded effectiveness value. Reporting that model through a risk–coverage curve places it in the selective-prediction framework, where a system may decline to answer in order to reduce error on what it does answer [10, 11]. Finally, a score can order failures well and still be a poor probability. The raw cross-encoder logit is not a probability, so our probability baseline is a Platt scaling of it [19] fitted on training data only, and calibration quality is measured with the Brier score [4] and reliability diagrams [12, 16]. Because we keep calibration separate from ranking quality, Section 4.4 can report an improvement and a degradation for the same model without contradiction.

Recent overlapping studies. Two recent studies overlap directly with this setting and are worth distinguishing carefully. Al-Joofi et al. [1] present a controlled empirical study of multi-stage retrieval for scientific literature that also combines lexical matching, dense retrieval, RRF fusion and cross-encoder reranking, and also evaluates on SciFact. Their emphasis is configuration-level: which pipeline composition performs best, with ablations on pseudo-relevance feedback and passage pooling, and cross-dataset checks on PubMedQA and SciDocs. They report cross-encoder reranking as the primary driver of their final gains and report an RRF "dilution" effect in which fusion does not consistently beat the strongest dense retriever. Our first-stage observation agrees with that second point, since fusion buys candidate recall at the cost of top-10 precision (Section 4.1), while our reranking result points the other way. The two studies are not directly comparable, because the encoders, candidate depths, indexing units and reported absolute numbers all differ. We therefore read the contrast as a reason to report reranker-specific evidence, not as a refutation of their finding. Apurba et al. [2] study retrieval and reranking for scientific retrieval-augmented generation over CORD-19 across corpus scales, and independently observe that an MS MARCO-trained cross-encoder reduces precision on a scientific corpus. Their retrieval labels are pseudo-relevance labels derived from their own hybrid system, and the work is a preprint that has not been peer reviewed, so we cite it as converging but weaker evidence.

How this study differs. This paper does not introduce a larger or better retrieval pipeline, and it makes no state-of-the-art claim. It focuses on the structural decomposition of failures, asking which stage could possibly have fixed a given failure, instead of ranking configurations. It keeps a held-out split that is evaluated exactly once under choices fixed in advance, treats failure discrimination and probability calibration as separate properties, and states which calibration-set findings replicated on held-out data. These are differences of protocol and framing on a single small dataset, not claims of novelty in retrieval modelling.

3 Experimental Design

3.1 Dataset and splits

SciFact is accessed through `ir_datasets` as `beir/scifact`: a corpus of 5,183 abstracts, 809 training claims (919 positive judgments) and 300 test claims (339 positive judgments). Every training and test claim used here has at least one positive relevance judgment.

The 809 training claims are partitioned 80/20 into 647 *calibration-train* and 162 *calibration-dev* queries by sorting the query identifiers and shuffling with `random.Random(42)`. Sorting before shuffling makes the partition independent of the dataset library's iteration order across versions. The resulting query-identifier files are committed to the repository, so later work reuses the stored split instead of recomputing it.

The 300-query test split was held out for the entire development of the study and opened exactly once, at the end, for the evaluation in Section 4.5. Sections 4.1–4.4 report calibration-set results, which is where every design decision was made.

3.2 Retrieval pipelines

Three first-stage retrievers each return the top 100 abstracts per query over corpus text formed as `title\n abstract`. **BM25** runs via `bm25s` [14] with parameters pinned explicitly ($k_1 = 1.5$, $b = 0.75$, $\delta = 0.5$, Lucene scoring and IDF, lower-casing, English stopwords, token pattern `(?u)\b\w\w+\b`, no stemmer); these were the library defaults at the time, and recording them concretely in each run manifest prevents a later release from silently changing the experiment. **Dense** uses `BAAI/bge-small-en-v1.5` [24] pinned to revision `5c38ec7`, with the model card's query instruction applied to queries only and cosine similarity over normalized embeddings. **Hybrid** is Reciprocal Rank Fusion [8] of the two rankings with $k = 60$, fixed in advance and never tuned, truncated back to 100 so all three pipelines emit the same candidate depth.

3.3 Reranking

All three pipelines are reranked by `cross-encoder/ms-marco-MiniLM-L6-v2` [3, 17], pinned to revision `c5ee24c`, over that pipeline's *own* top 50 candidates. No model is fine-tuned. The reranker can only reorder what its retriever supplied, and that constraint makes the decomposition below well defined.

3.4 Failure taxonomy

For a query q with gold abstracts $G(q)$, first-stage ranking $R_1(q)$ and reranked ranking $R_2(q)$:

$$\text{candidate\_success}_{50}(q) = \mathbf{1}[\, G(q) \cap R_1(q)_{1:50} \neq \emptyset \,], \tag{1}$$

$$\text{final\_success}_{10}(q) = \mathbf{1}[\, G(q) \cap R_2(q)_{1:10} \neq \emptyset \,]. \tag{2}$$

Each query then receives exactly one of five transition labels. The candidate check is evaluated first and is the only exit for $\text{candidate\_success}_{50} = 0$, so a candidate-set failure can never be mislabelled as a reranker rescue.

Label	**Condition**
`no_opportunity`	gold not in the top-50 candidate set
`already_successful`	gold in the top 10 both before and after reranking
`rescued_by_reranker`	gold absent from the first-stage top 10, present after
`degraded_by_reranker`	gold present in the first-stage top 10, absent after
`unchanged_failure`	gold in the candidate set but in neither top 10

The **candidate-set failure rate** is the share of all queries labelled `no_opportunity`; the **reranking failure rate** is the share labelled `degraded_by_reranker` or `unchanged_failure`; the **final success rate** is the share labelled `already_successful` or `rescued_by_reranker`. These three quantities partition the query set exactly. Two further quantities are named for their denominators: *success given opportunity* is final success among queries that had a candidate-set opportunity, and the *share of failures from the candidate set* is the fraction of final failures the reranker never had a chance to fix.

3.5 Failure-risk prediction

Eight features are computed per query, all from the pipeline's own output and none from relevance judgments. Four are first-stage: the top-1 score and the top1–top2 margin, each min–max normalized within the query's own 50-candidate set; the Spearman correlation between first-stage and reranked positions; and whether the first-stage top-1 document survives into the reranked top 3. Within-query normalization lets one feature definition serve BM25 scores, cosine similarities and RRF fusion scores without a fitted cross-query transform. Four are reranker-side: the top-1 score, the top1–top2 margin, and the mean and standard deviation of the top-5 reranker scores.

A `StandardScaler` followed by `LogisticRegression` with `class_weight="balanced"` predicts $\text{final\_success}_{10}$. The class weighting was chosen from the 83/17 class imbalance measured on calibration-train, before any development-set result was seen. Two baselines are reported beside the risk model and kept separate from it: the **raw reranker top-1 score**, used for ranking metrics only, since it is not a probability and so no Brier score is computed for it; and that same score after **Platt scaling fitted on calibration-train** [19], which gives the baseline a genuine probability and therefore a like-for-like Brier score. Platt scaling is monotone, so it leaves AUROC and AUPRC unchanged. A hybrid-only exploratory ablation adding the BM25/dense top-10 overlap feature is reported alongside the primary model, and never replaces it.

3.6 Statistical analysis

All comparisons use a paired query-level percentile bootstrap: 1,000 resamples, 95% intervals, seed 42, with the same resampled query identifiers applied to both sides and the difference always reported as $B - A$. Resamples that become single-class fall back to the degenerate-but-defined AUROC/AUPRC value instead of being dropped, which would bias the interval toward whichever side survives more often. Intervals are reported for a fixed, pre-specified list of comparisons, not for every possible pair. We describe an interval that excludes zero as *significant* in this percentile-bootstrap sense; no multiplicity correction is applied across the comparison list, which is a further reason to read individual intervals conservatively.

The pre-specified protocol evaluates the risk model on the 162 calibration-dev queries, which contain only about 20 failures, too few to separate the models. A secondary, higher-powered estimate therefore pools out-of-fold predictions from stratified 5-fold cross-validation over all 809 calibration queries, raising the failure count to 117–137 depending on pipeline. Within each fold the scaler, the logistic regression and the Platt baseline are fitted on that fold's training portion alone. This changes estimation power, not model selection: thresholds are still chosen on calibration-dev by the pre-specified rule, the train/dev result is still reported unchanged, and the test split remains untouched.

3.7 Leakage controls

The dataset, splits, model identities, candidate depth, failure definitions, primary metrics, the confidence target, the calibration method and the list of statistical comparisons were fixed in a project protocol document written before any experiment was executed, and the dated decisions taken under it are recorded in an experiment log released with the code. We do not call this a preregistration, since no public, externally verifiable preregistration exists. The protocol was documented before the experiments ran, and the analysis choices were recorded before the held-out evaluation.

The operational controls are as follows. The scaler and the logistic regression are fitted on calibration-train only; abstention thresholds are selected on calibration-dev only; no feature is derived from relevance judgments; and RRF was never tuned. The primary pipeline for the risk-model analysis was chosen by a rule fixed in advance, the highest calibration-dev Recall@50, which picked the hybrid pipeline. Reaching the held-out split requires calling a separately named loader: the calibration split resolver accepts only the two calibration split names and raises on anything else, including `test`, and unit tests assert this, that no development-set feature value ever reaches a `fit` call, that cross-validation predicts each query exactly once out of fold, and that the raw reranker score is never reported as a probability. Opening the held-out data therefore took an explicit, greppable change to the code.

4 Results

4.1 First-stage retrieval

Table 1 reports the three first stages on both splits. Fusion buys candidate recall at the cost of top-10 precision: the hybrid pipeline leads on Recall@50 by 1.8 points over dense on calibration data and by 1.9 points held out, while trailing dense on nDCG@10 on both splits. Under the pre-specified selection rule, highest calibration-dev Recall@50, hybrid is the primary pipeline for the risk-model analysis, and the same rule picks the same pipeline out of sample.

4.2 Candidate-set versus reranking failures

Table 2 and Figure 1 give the central result. Moving from BM25 to hybrid retrieval cuts candidate-set failures by close to a factor of four on calibration data, from 11.7% to 3.1% of all queries, yet final success improves by only 3.1 points, from 84.6% to 87.7%. The reason is visible in the last column: the failures changed category instead of disappearing. Three quarters of BM25's failures are queries the reranker could not have fixed; only a quarter of the hybrid pipeline's are. Reranking failures rise correspondingly from 3.7% to 9.3% of all queries, as more borderline evidence reaches the reranker and is not ranked into the top ten.

For this dataset and these models, further first-stage work has little headroom left. The hybrid

Table 1: First-stage retrieval before reranking; best value per column and panel in bold. Fusion leads on Recall@50 on both splits while trailing dense on nDCG@10. Held-out values are 2–5 points lower throughout, as expected for a split never used for any decision.

Pipeline	Recall@5	Recall@10	Recall@50	MRR@10	nDCG@10
Calibration-dev ($n = 162$)					
BM25	0.752	0.824	0.873	0.684	0.715
Dense (BGE)	0.826	**0.866**	0.948	**0.744**	**0.769**
Hybrid (RRF)	**0.828**	0.864	**0.966**	0.718	0.751
Held-out test ($n = 300$)					
BM25	0.719	0.774	0.869	0.631	0.662
Dense (BGE)	**0.759**	**0.836**	0.925	**0.682**	**0.713**
Hybrid (RRF)	0.758	0.822	**0.944**	0.667	0.700

Table 2: Failure decomposition. The first three columns partition the query set exactly and sum to 100% per row. "Success given opportunity" conditions on the gold abstract reaching the candidate set; "failures from candidate set" is the share of *final failures* the reranker never had a chance to fix.

Pipeline	Candidate-set failure	Reranking failure	Final success	Success given opportunity	Failures from candidate set
Calibration-dev ($n = 162$)					
BM25	11.7%	3.7%	84.6%	95.8%	76.0%
Dense (BGE)	4.9%	8.0%	87.0%	91.6%	38.1%
Hybrid (RRF)	3.1%	9.3%	**87.7%**	90.4%	25.0%
Held-out test ($n = 300$)					
BM25	12.3%	6.0%	81.7%	93.2%	67.3%
Dense (BGE)	7.0%	9.3%	83.7%	90.0%	42.9%
Hybrid (RRF)	5.3%	10.3%	**84.3%**	89.1%	34.0%

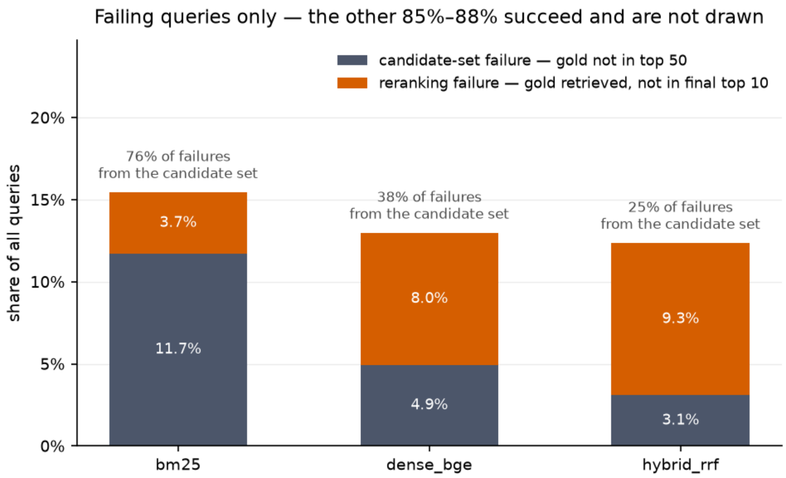

Figure 1: Failing queries only, on calibration-dev; the successful majority is omitted so the composition is visible. As candidate recall improves from BM25 to hybrid fusion, the total bar shrinks modestly while its composition inverts.

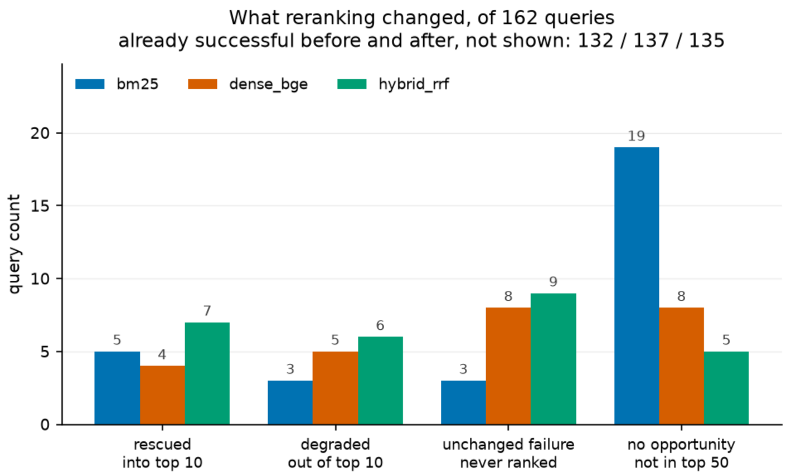

Figure 2: Transition counts on calibration-dev; queries already successful before and after reranking (132, 137 and 135 of 162) are omitted. Rescues and degradations are of comparable magnitude for every pipeline, which is why the aggregate metrics in Table 3 barely move.

Table 3: Effect of reranking: paired query-level differences (after − before) with 95% percentile bootstrap intervals (1,000 resamples, seed 42). Bold marks the single interval excluding zero; it did not replicate held out.

	Recall@10		nDCG@10	
Pipeline	Δ	95% CI	Δ	95% CI
Calibration-dev ($n = 162$)				
BM25	+0.015	[−0.019, +0.049]	−0.001	[−0.039, +0.036]
Dense (BGE)	+0.001	[−0.037, +0.038]	**−0.044**	[**−0.081**, **−0.009**]
Hybrid (RRF)	+0.006	[−0.040, +0.049]	−0.026	[−0.062, +0.010]
Held-out test ($n = 300$)				
BM25	+0.027	[−0.005, +0.060]	+0.022	[−0.008, +0.052]
Dense (BGE)	−0.013	[−0.043, +0.018]	−0.017	[−0.043, +0.012]
Hybrid (RRF)	+0.007	[−0.030, +0.041]	−0.004	[−0.034, +0.024]

pipeline already retrieves annotated evidence into the top 50 for 96.9% of calibration queries and 94.7% of held-out queries, so the binding constraint has moved to the reranking stage.

4.3 Effect of reranking

Figure 2 shows what reranking actually changed on calibration-dev. The reranker rescues 5, 4 and 7 queries into the top ten for BM25, dense and hybrid respectively, and pushes 3, 5 and 6 out of it. Those counts are close to a wash.

Table 3 gives the paired before/after differences with bootstrap intervals. Recall@10 changes are small and every interval crosses zero on both splits. On calibration data nDCG@10 falls for all three pipelines and the dense interval lies entirely below zero; on held-out data no before/after comparison is significant for any pipeline, and BM25 nDCG@10 actually rises. The mechanism behind the calibration-set drop is visible in the transition counts: a rescue typically moves a document into a mid-top-10 position, whereas a degradation removes a document that was already ranked highly, so degradations cost more nDCG than rescues recover even when the counts balance.

The claim this study supports is therefore the weaker one: for the evaluated MS MARCO cross-encoder on SciFact, reranking does not reliably improve final ranking quality, and the stronger statement that it degrades dense retrieval is not supported once held-out data is taken into

Table 4: Failure-risk prediction. *Top panel*: cross-validated over all 809 calibration queries (pooled out-of-fold predictions from stratified 5-fold cross-validation; 117–137 failures depending on pipeline). *Bottom panel*: the single held-out evaluation (300 queries, roughly 50 failures), with models fitted on calibration-train and never refitted. Differences are model − baseline with 95% paired bootstrap intervals; AUROC and AUPRC compare against the raw reranker score, Brier against the train-fitted Platt scaling of that score. Higher is better for AUROC and AUPRC; *lower* is better for Brier, so the Brier rows are unfavourable to the risk model. Bold marks intervals excluding zero.

Pipeline	Metric	Baseline	Risk model	Difference	95% CI
Cross-validated over calibration queries ($n = 809$)					
BM25	AUROC	0.781	0.846	**+0.065**	[**+0.034**, **+0.098**]
Dense (BGE)	AUROC	0.757	0.840	**+0.083**	[**+0.039**, **+0.124**]
Hybrid (RRF)	AUROC	0.758	0.827	**+0.069**	[**+0.034**, **+0.105**]
BM25	AUPRC	0.936	0.962	**+0.025**	[**+0.011**, **+0.043**]
Dense (BGE)	AUPRC	0.940	0.965	**+0.025**	[**+0.010**, **+0.042**]
Hybrid (RRF)	AUPRC	0.939	0.962	**+0.023**	[**+0.009**, **+0.039**]
BM25	Brier	0.115	0.163	**+0.048**	[**+0.031**, **+0.065**]
Dense (BGE)	Brier	0.110	0.162	**+0.052**	[**+0.032**, **+0.071**]
Hybrid (RRF)	Brier	0.113	0.171	**+0.058**	[**+0.039**, **+0.076**]
Held-out test split ($n = 300$), evaluated once					
BM25	AUROC	0.767	0.811	+0.044	[−0.005, +0.092]
Dense (BGE)	AUROC	0.783	0.831	+0.049	[−0.017, +0.120]
Hybrid (RRF)	AUROC	0.750	0.784	+0.034	[−0.028, +0.102]
BM25	AUPRC	0.937	0.951	+0.014	[−0.003, +0.032]
Dense (BGE)	AUPRC	0.947	0.960	+0.014	[−0.007, +0.036]
Hybrid (RRF)	AUPRC	0.942	0.949	+0.007	[−0.012, +0.027]
BM25	Brier	0.131	0.179	**+0.048**	[**+0.018**, **+0.078**]
Dense (BGE)	Brier	0.113	0.166	**+0.054**	[**+0.024**, **+0.084**]
Hybrid (RRF)	Brier	0.119	0.194	**+0.075**	[**+0.043**, **+0.107**]

account. This is a transfer result about one general-domain reranker applied unmodified to scientific claims, and it says nothing about cross-encoders as a class. The converging observation of Apurba et al. [2] on a different scientific corpus is likewise about transfer.

4.4 Failure-risk prediction

Under the pre-specified train/dev protocol, every raw-versus-model interval crossed zero: with roughly 20 failures among 162 queries, the comparison was underpowered rather than null. On calibration-dev the risk model reached AUROC 0.851 against 0.804 for the raw score on the hybrid pipeline, with a Brier score of 0.166 against 0.097 for the Platt-scaled baseline. The cross-validated estimate over 809 calibration queries (Table 4) resolves the comparison in both directions.

The risk model discriminates better on all three pipelines and is calibrated worse on all three. AUPRC deserves a caveat: it is bounded below by the positive class rate, which is 0.852 on the pooled hybrid data, not 0.5. Measured against that floor, the raw score captures 0.088 and the risk model 0.110, so the +0.023 difference is roughly a 25% relative improvement in the informative part of AUPRC, not the ∼2% the bare numbers suggest. The released artifacts report `base_rate` and `auprc_over_base_rate` for this reason.

Figure 3 shows the mechanism. The risk model's points lie above the diagonal at nearly every bin: it systematically understates the probability of success. The same wider spread of predictions explains both results: more separation between queries improves the ranking metrics, and pushing predictions toward the extremes costs the model on Brier.

Class-weight ablation. `class_weight="balanced"` was fixed in advance from the training-split class imbalance and was never revised, so the model reported above is the pre-specified one

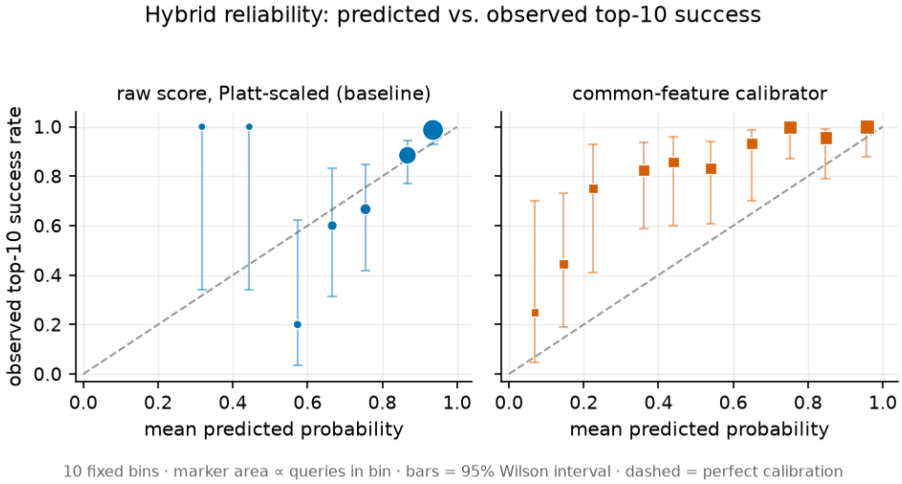

Figure 3: Reliability for the hybrid pipeline on calibration-dev. The panel labelled *common-feature calibrator* is the failure-risk model of Section 3.5; the artifact name is kept for traceability. Its points lie above the diagonal in nearly every bin, so it systematically understates success probability; the Platt-scaled baseline never predicts below 0.3, leaving its first three bins empty.

Table 5: Post-hoc class-weight ablation, cross-validated over all 809 calibration queries. Differences are unweighted − comparator with 95% paired bootstrap intervals (1,000 resamples, seed 42). Brier is a loss, so negative is better; AUROC is a score, so a negative value there would mean discrimination was given up. Bold marks intervals excluding zero.

	Brier		Δ Brier vs.		Δ AUROC
Pipeline	pre-spec.	unweighted	pre-specified	Platt	vs. pre-specified
BM25	0.163	0.103	**−0.060** [**−0.074**, **−0.046**]	**−0.012** [**−0.019**, **−0.005**]	+0.001 [−0.001, +0.004]
Dense (BGE)	0.162	0.096	**−0.066** [**−0.081**, **−0.050**]	**−0.015** [**−0.023**, **−0.006**]	+0.002 [−0.002, +0.005]
Hybrid (RRF)	0.171	0.102	**−0.070** [**−0.084**, **−0.054**]	**−0.012** [**−0.019**, **−0.004**]	−0.001 [−0.003, +0.002]

throughout. To test whether that setting is what costs the model its probability quality, we refit the same features without it — on the same folds, with the same seed — after the result above had been observed. Table 5 reports it as a post-hoc ablation, not as a competing configuration. Removing the weighting improves Brier on every pipeline with every interval excluding zero, while leaving AUROC unchanged with every interval crossing zero; the unweighted refit is also better calibrated than the Platt baseline the pre-specified model lost to. The calibration deficit therefore belongs to the class weighting rather than to the feature set, and the weighting bought no discrimination in exchange. Because the held-out split was spent before this ablation existed (Section 4.5), the unweighted model has no out-of-sample evaluation: it explains the pre-specified result and does not replace it.

The model is therefore usable for *ordering* queries by risk (routing, abstention, triage), but not for reporting a probability to a user. Figure 4 makes the operating range concrete: on calibration-dev, answering only the 60% of queries the risk model is most confident about raises success from 87.7% to 97.9%, against 94.8% at the same coverage under the raw score, while at 80% coverage the two are within a point of each other (93.8% versus 94.6%). The cross-validated pool shows the same pattern with more queries behind it: 95.9% against 93.0% at 60% coverage.

4.5 Held-out evaluation

The test split was opened once, after Sections 4.1–4.4 were complete and every model, feature and threshold was fixed. The risk models are the calibration-train models used unchanged; the absten-

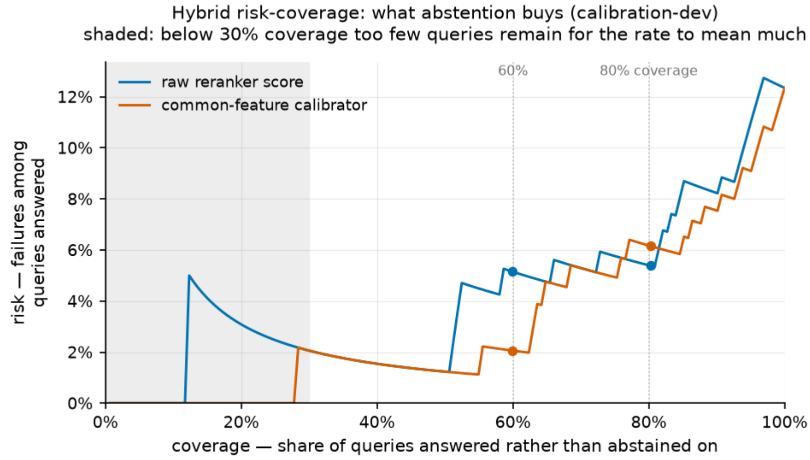

Figure 4: Risk–coverage for the hybrid pipeline on calibration-dev. Risk is the failure rate among answered queries; coverage is the share of queries answered instead of abstained on. The shaded region (below 30% coverage) holds too few queries for the rate to be informative. The risk model's advantage appears where abstention is aggressive.

Table 6: Replication summary. Where a held-out result contradicted a calibration-set result, the calibration-set claim was weakened rather than the reverse.

Finding	Calibration	Held-out	Replicated?
Best Recall@50	hybrid, 0.966	hybrid, 0.944	Yes; selection rule holds out of sample
Share of failures from candidate set	76% → 25%	67% → 34%	Yes; same ordering and direction
Hybrid vs. BM25 final success	+0.031, CI touches 0	+0.027, CI [+0.003, +0.050]	Yes, and stronger held out
Reranking degrades dense nDCG@10	−0.044, CI [−0.081, −0.009]	−0.017, CI [−0.043, +0.012]	**No**
Risk-model Brier worse than Platt	+0.048 to +0.058 (CV)	+0.048 to +0.075	Yes; significant on all three
Risk-model AUROC better than raw	+0.065 to +0.083 (CV)	+0.034 to +0.049, CIs cross 0	Direction yes, power no

tion thresholds are those selected on calibration-dev. Nothing was refitted and no configuration was altered in response to any result below.

The lower panel of Table 4 reports the held-out risk-model metrics and Table 6 summarizes which calibration-set findings replicated. Four points stand out.

First, the primary-pipeline rule survives out of sample: hybrid again has the highest Recall@50 (Table 1). Second, the composition shift replicates with the same ordering and a smaller spread (Table 2). Third, the pipeline comparison is *stronger* held out: the hybrid pipeline's final success exceeds BM25's by +0.027 with a 95% interval of [+0.003, +0.050], where the corresponding calibration interval had merely touched zero (+0.031, [+0.000, +0.068]). Fourth, the significant dense reranking degradation of Section 4.3 did not replicate.

For the risk model, the calibration failure replicates decisively, with the Brier degradation significant on every pipeline and every split tested, while the discrimination advantage replicates only in direction. All three held-out AUROC differences are positive and none is individually significant at 300 queries holding roughly 50 failures; this is the same power limitation that made the pre-specified calibration-dev comparison inconclusive, and the reason the cross-validated estimate exists. Held-out data agrees in direction with that estimate without being able to confirm it on its own.

Table 7: Threshold transfer for the hybrid pipeline. Each cutoff was selected on calibration-dev to hit the target coverage, then applied unchanged to the 300 held-out queries; "achieved" is the share of held-out queries kept.

	Raw reranker score		Failure-risk model	
Target coverage	Achieved	Success rate	Achieved	Success rate
100%	99.7%	0.846	98.0%	0.857
80%	74.0%	0.892	71.7%	0.916
60%	48.3%	0.952	58.0%	0.931

4.6 Threshold transfer

Held-out data also answers an operational question the calibration split cannot: whether an abstention threshold chosen on calibration-dev still delivers its intended coverage on unseen queries. Table 7 carries the hybrid thresholds over unchanged.

At the aggressive setting the risk model's cutoff transferred more consistently: it landed within two points of its 60% target where the raw score undershot by roughly twelve points. The narrow reading is the right one. The risk model's *score distribution*, and with it the selected cutoff, kept its shape better across these two splits. In absolute terms its probabilities remain poorly calibrated, as the Brier results in Table 4 show. Threshold transfer and probability calibration are different properties, and only the former is supported here. It is also a single observation on one split, at one operating point, for one pipeline.

5 Qualitative Failure Analysis

Twelve hybrid calibration-dev queries were selected by fixed rules before any case text was read: the two highest-confidence incorrect queries first, then evenly spaced confidence quantiles within each transition category, with query identifier as a stable tie-break. The selection is recorded as a machine-readable artifact so that it cannot be retrofitted to the narrative. The counts below describe those twelve cases and are not estimates of population frequency.

Four failure modes recur. **Evidence outside the cutoff** (7 of 12 cases) is the most common: the annotated abstract exists in the corpus and is topically adjacent but never enters the top 50. **Terminology mismatch** (4) covers claims phrased in one register, such as "increased flux of microbial products", whose annotated evidence is written in another, here in terms of intestinal barrier compromise. **Lexical distraction** (4) covers claims combining two concepts where one dominates retrieval: a claim about ischemia-reperfusion *in aging* retrieves ischemia papers and misses the aging study. **Cross-encoder overconfidence** (3) covers the reranker's preference for near-verbatim title matches. In the single highest-confidence error, an abstract naming "B cells", "exhaustion" and "HIV" in its title is scored 3.97 while every other candidate falls below zero, and the annotated review, which discusses antibody responses more broadly, stays below the cutoff.

The two rescues are informative in the opposite direction. In one, the first stage is distracted by generic inhibitor language and the reranker recognises the exact compound and mechanism strongly enough to pull the annotated document into the top ten. Pairwise scoring does add something on this data, but it gives back about as much as it gains.

A caveat applies throughout: SciFact marks *annotated* evidence, not the complete set of scientifically relevant abstracts. Several cases counted here as failures are abstracts a domain reader might well call relevant but which were not the annotated source. The risk model therefore predicts retrieval of annotated evidence, not scientific correctness, and the rates in Table 2 should be read the same way.

6 Discussion

The decomposition changes what an aggregate number means. Read only through final success, the hybrid pipeline is a three-point improvement over BM25 on calibration data and a 2.6-point improvement held out; read through the decomposition, it nearly eliminates one failure mode while a second grows to partially replace it. A practitioner who saw only the aggregate might invest in a better first stage; the decomposition says that on SciFact, with these models, that investment has little headroom left.

That reading depends on the second result. Under the evaluated frozen models, the standard second stage does not reliably improve final ranking quality here: rescues and degradations are of comparable magnitude, and no held-out before/after comparison is significant. Applying a general-domain MS MARCO cross-encoder unmodified to scientific claims carries a cost in this setting. This is consistent with BEIR's observation that no single paradigm dominates every collection [22], and it stands in apparent tension with Al-Joofi et al. [1], who report reranking as the primary driver of gains on SciFact under a different pipeline configuration. Resolving that tension would require holding the reranker, the candidate depth and the indexing unit fixed across both setups, which we have not done.

The third result concerns what "confidence" means operationally. One lightweight model is better than the raw reranker score at ordering queries by risk and worse than a one-parameter Platt baseline at stating the probability that a query will succeed; both directions are significant in the cross-validated estimate, and the calibration degradation is significant held out as well. For a routing or abstention system, which consumes an ordering and a cutoff, that trade is often acceptable; for anything that displays a confidence number to a user, it is not. Because the class weighting responsible for the compression was fixed in advance and never revised, the model we report is the pre-specified one; the ablation in Section 4.4 then shows that the trade was avoidable, since removing that weighting recovers calibration without costing discrimination. What we would expect to carry over is therefore not the trade itself but the reason for measuring the two properties apart: reported alone, AUROC would have hidden the defect and Brier would have made its remedy look like a loss.

Finally, the protocol is part of what the study offers. Because the failure definitions, leakage rules, metrics and comparison list were all fixed before any run existed, the non-replication in Section 4.5 is reportable instead of embarrassing: the dense reranking comparison was specified in advance, so its calibration-set result could not be quietly dropped once held-out data disagreed.

7 Limitations

One dataset, and a small one. SciFact has 5,183 abstracts, 162 calibration-dev queries and 300 held-out queries. Nothing here establishes that the composition shift, or any other result, generalizes to other collections, domains or corpus sizes. Every claim is scoped to SciFact under the evaluated models.

Small failure counts. Roughly 20 failures in the development split made the pre-specified risk-model comparison inconclusive and motivated the cross-validated secondary analysis; roughly 50 held-out failures are why the held-out AUROC comparison is directionally consistent but not individually significant. Development-only intervals should be read as underpowered rather than as evidence of no effect, and no multiplicity correction is applied across the pre-specified comparison list.

Off-the-shelf, frozen models. The reranking result is evidence about MS MARCO $\rightarrow$ SciFact transfer for one specific cross-encoder, not about cross-encoders in general; a domain-adapted reranker might behave very differently. RRF likewise ran at $k = 60$ untuned, so the hybrid pipeline is a fixed comparator, not an optimized system.

One risk-model configuration, and one unvalidated alternative. The Brier degradation is attributable to the pre-specified class weighting, measured by the post-hoc ablation in Section 4.4. That ablation is calibration-data only: the held-out split had been spent before it existed, so the better-calibrated unweighted model has never been evaluated out of sample and cannot be recommended on the strength of these results alone. No other hyperparameter of the risk model was varied.

Annotated evidence is not scientific truth. A topically excellent non-annotated abstract counts as a failure in these tables, and Section 5 shows concrete instances. The risk model's target is retrieval of annotated evidence, not the correctness of a claim.

The held-out split is spent. It was evaluated once, as designed, and will not be used again, so no untouched replication set remains within SciFact. A future change to this pipeline would need a new dataset, or a new split protocol, to be validated comparably.

One claim was weakened by held-out data. The dense reranking degradation is reported in Section 4.3 because that is what the calibration data showed and because the protocol fixed that comparison in advance; Section 4.5 records that it did not replicate. The calibration-set effect sizes throughout should be read as the optimistic end of the range.

8 Conclusion

Decomposing retrieval failure changes what aggregate metrics mean. On SciFact, under frozen general-domain models, hybrid fusion looks like a modest three-point improvement in final top-10 success over BM25; the decomposition shows that it nearly eliminates candidate-set failures while reranking failures grow to partially replace them, leaving the reranking stage as the binding constraint. That shift replicates on a held-out split evaluated once. For the evaluated MS MARCO cross-encoder, reranking does not reliably improve final ranking quality here. Failure also turns out to be predictable from retrieval-internal signals alone: a lightweight risk model orders at-risk queries better than the reranker's own score and carries an abstention cutoff across splits more consistently, while stating significantly worse-calibrated probabilities than a one-parameter Platt baseline. A post-hoc ablation locates that miscalibration in the pre-specified class weighting rather than in the features: refitting without it recovers calibration at unchanged discrimination, though a spent test split leaves that refit unvalidated out of sample.

Two of these results are negative. A fourth, that reranking significantly degrades dense retrieval, appeared on calibration data and did not survive held-out evaluation; we report it as not replicating instead of dropping it quietly. The findings belong to SciFact and to the models evaluated here. What we would expect to carry over is the diagnostic itself, not its numbers.

9 Reproducibility Statement

The code, pipeline configurations, committed split files, result tables, figures and run manifests underlying every number in this paper are maintained in the project repository. The repository was not public at the time of writing; the URL will be stated once it is released or archived. Tables and figures are regenerated from saved run artifacts by a single command; no number in this manuscript was computed by hand, and each was checked against the corresponding committed JSON or Markdown result file.

Model identities and revisions are pinned where the artifacts support it: the dense encoder at revision `5c38ec7` and the cross-encoder at `c5ee24c` (full 40-character hashes are recorded in every run manifest), with BM25 parameters recorded explicitly in each run manifest. Every committed result traces to a run directory, a Git commit hash and a model revision through the stored manifests; the manifests also record whether the working tree was clean at run time, and the calibration runs backing Sections 4.1–4.4 were executed from a clean tree. Per-query run directories, cached embeddings and model weights are not committed, as they are regenerable; the commands to regenerate them are documented in the repository.

The held-out test split was evaluated exactly once, after the retrieval implementations, configurations, features, models, thresholds and the primary-pipeline selection rule were all fixed, and nothing was refitted or altered in response to the result. Consequently the SciFact test split is now *spent* for this line of work: it can no longer serve as untouched validation data for future modifications of this system, and no independent replication set remains available within SciFact.

The protocol document that fixed the dataset, splits, models, failure taxonomy, leakage rules and statistical comparisons was written before any experiment was executed, and the dated decisions taken under it, including those made after seeing unfavourable results, are recorded in the repository's experiment log. This is a documented internal protocol and not a public preregistration.

References

[1] Walaa Al-Joofi, Alaa Sagheer, and Hala Hamdoun. A multi-stage hybrid retrieval framework for the scientific literature with cross-encoder re-ranking. *Applied Sciences*, 16(10):4813, 2026. doi: 10.3390/app16104813.

[2] Kaysarul Anas Apurba, Md. Hasibul Hasan, Rofiqul Alam Shehab, and Asab Azad. SciRet: A compute-aware empirical study of retrieval and reranking for scientific RAG, 2026. URL https://arxiv.org/abs/2608.03860. arXiv:2608.03860; preprint, no peer-reviewed publication recorded as of 2026-08-12.

[3] Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. MS MARCO: A human generated machine reading comprehension dataset, 2016. URL https://arxiv.org/abs/1611.09268. arXiv:1611.09268.

[4] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

[5] David Carmel and Elad Yom-Tov. *Estimating the Query Difficulty for Information Retrieval*. Synthesis Lectures on Information Concepts, Retrieval, and Services. Morgan & Claypool, 2010. doi: 10.2200/S00235ED1V01Y201004ICR015.

[6] Adrian-Gabriel Chifu, Sébastien Déjean, Josiane Mothe, Moncef Garouani, Diego Ortiz, and Md Zia Ullah. Uncovering the limitations of query performance prediction: Failures, insights, and implications for selective query processing, 2025. URL https://arxiv.org/abs/2504.01101. arXiv:2504.01101; preprint, no journal or conference publication recorded as of 2026-08-12.

[7] Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel S. Weld. SPECTER: Document-level representation learning using citation-informed transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, 2020. doi: 10.18653/v1/2020.acl-main.207. URL https://aclanthology.org/2020.acl-main.207/.

[8] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009. doi: 10.1145/1571941.1572114.

[9] Steve Cronen-Townsend, Yun Zhou, and W. Bruce Croft. Predicting query performance. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 299–306, 2002. doi: 10.1145/564376.564429.

[10] Ran El-Yaniv and Yair Wiener. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11:1605–1641, 2010. URL https://www.jmlr.org/papers/v11/el-yaniv10a.html.

[11] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, pages 4878–4887, 2017. URL https://arxiv.org/abs/1705.08500. arXiv:1705.08500.

[12] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

[13] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781, 2020. doi: 10.18653/v1/2020.emnlp-main.550. URL https://aclanthology.org/2020.emnlp-main.550/.

[14] Xing Han Lù. BM25S: Orders of magnitude faster lexical search via eager sparse scoring, 2024. URL https://arxiv.org/abs/2407.03618. arXiv:2407.03618.

[15] Chuan Meng, Negar Arabzadeh, Mohammad Aliannejadi, and Maarten de Rijke. Query performance prediction: From ad-hoc to conversational search. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2583–2593, 2023. doi: 10.1145/3539618.3591919.

[16] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005. doi: 10.1145/1102351.1102430.

[17] Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT, 2019. URL https://arxiv.org/abs/1901.04085. arXiv:1901.04085.

[18] Rodrigo Nogueira, Zhiying Jiang, Ronak Pradeep, and Jimmy Lin. Document ranking with a pretrained sequence-to-sequence model. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 708–718, 2020. doi: 10.18653/v1/2020.findings-emnlp.63. URL https://aclanthology.org/2020.findings-emnlp.63/.

[19] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

[20] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 4(1–2):1–174, 2009. doi: 10.1561/1500000019.

[21] Anna Shtok, Oren Kurland, David Carmel, Fiana Raiber, and Gad Markovits. Predicting query performance by query-drift estimation. *ACM Transactions on Information Systems*, 30(2):1–35, 2012. doi: 10.1145/2180868.2180873.

[22] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogenous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021. URL https://arxiv.org/abs/2104.08663. arXiv:2104.08663.

[23] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.609. URL https://aclanthology.org/2020.emnlp-main.609/.

[24] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-Pack: Packed resources for general Chinese embeddings. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 641–649, 2024. doi: 10.1145/3626772.3657878.